

14. *Assertion (A) :* The cylindrical soft iron core in a moving coil galvanometer only makes the magnetic field radial and does not affect the strength of the magnetic field.

Reason (R) : In a moving coil galvanometer, the plane of the coil is always perpendicular to the magnetic field.

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Reason (R) : In a moving coil galvanometer, the plane of the coil is always perpendicular to the magnetic field.

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(D) Both Assertion (A) and Reason (R) are false

10. A galvanometer of resistance $27\ \Omega$ is converted into an ammeter of range (0 – 10 mA) using a resistance of $3\ \Omega$. The galvanometer will show full scale deflection for a current of about –

(A) 10 mA

(B) 100 mA

(C) 1 mA

(D) 3 mA

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1

(C) 1 mA

29. A galvanometer is used to detect or/and measure small currents in an electrical circuit. It essentially works on the fact that a current-carrying coil experiences a deflecting torque when placed in a magnetic field. This deflection in the coil can be measured and it is related to the current flowing in the coil, the number of turns in the coil, area of the coil and the magnetic field. A hair spring attached to the coil provides a counter torque and helps in measuring the deflection. A galvanometer can be converted to an ammeter or a voltmeter of desired range by using suitable resistances.

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(I) The torque on the coil remains constant irrespective of the coil's orientation during rotation due to

- (A) use of soft iron core which increases the magnetic field.
- (B) radial magnetic field
- (C) hair spring which provides the counter torque
- (D) eddy current in the iron core which causes damping.

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(B) radial magnetic field

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(II) The best way to increase current sensitivity of a galvanometer is by

- (A) increasing number of turns of the coil
- (B) increasing area of coil and magnetic field strength
- (C) decreasing area of coil and magnetic field strength
- (D) increasing torsional constant of the hair spring

(II) The best way to increase current sensitivity of a galvanometer is by

(A) increasing number of turns of the coil

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(C) decreasing area of coil and magnetic field strength

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(III) A moving coil galvanometer has a coil with area of cross-section $4.0 \times 10^{-3} \text{ m}^2$ and number of turns 50. The coil is rotating in a magnetic field of 0.25 T. The torque acting on the coil when a current of 5 A passes through it is

(A) 1.0 N m

(B) 2.0 N m

(C) 0.50 N m

(D) 0.25 N m

(III) A moving coil galvanometer has a coil with area of cross-section $4.0 \times 10^{-3} \text{ m}^2$ and number of turns 50. The coil is rotating in a magnetic field of 0.25 T. The torque acting on the coil when a current of 5 A passes through it is

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A galvanometer coil has a resistance of $15\ \Omega$ and the meter shows full scale deflection for a current of $3\ \text{mA}$. The value of resistance required to convert it into a voltmeter of range $(0 - 12\ \text{V})$ is

(A) $4015\ \Omega$

(B) $3985\ \Omega$

(C) $415\ \Omega$

(D) $385\ \Omega$

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(IV) A galvanometer with coil of resistance $20\ \Omega$ shows full scale deflection for a current of $5\ \text{mA}$. To convert it into an ammeter of range $(0 - 10\ \text{A})$, a resistance of

- (A) $0.05\ \Omega$ should be connected in series with it.
- (B) $0.05\ \Omega$ should be connected in parallel with it.
- (C) $0.01\ \Omega$ should be connected in parallel with it.
- (D) $0.01\ \Omega$ should be connected in series with it.

(IV) A galvanometer with coil of resistance $20\ \Omega$ shows full scale deflection for a current of $5\ \text{mA}$. To convert it into an ammeter of range $(0 - 10\ \text{A})$, a resistance of

(A) $0.05\ \Omega$ should be connected in series with it.

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(C) $0.01\ \Omega$ should be connected in parallel with it.

(D) $0.01\ \Omega$ should be connected in series with it.

- 33.** (a) (i) Write the principle and explain the working of a moving coil galvanometer. A galvanometer as such cannot be used to measure the current in a circuit. Why ?
- (ii) Why is the magnetic field made radial in a moving coil galvanometer ? How is it achieved ?

5

(i) For a moving coil galvanometer	
Principle	1
Working	1
Reason it cannot be used as such	1
(ii) Reason for radial field	1
How radial field is achieved	1

(i) Principle – When a rectangular loop carrying current I is placed in a uniform magnetic field, it experiences a torque. Working:- When a current flows through the coil of a galvanometer, a torque acts on it. $\tau = NiAB \sin \theta$ For radial magnetic field; $\sin \theta = 1$ The spring provides a counter or restoring toqrue $k\phi$. $k\phi = NiAB$ In equilibrium; $\phi = \left(\frac{NAB}{k} \right) i$ Galvanometer cannot be used as such to measure current because: -It has large resistance and hence will change the value of current in the circuit. -It is a sensitive device. (Any one of the above)	1
(ii) The magnetic field is made radial in a moving coil galvanometer so that the magnetic dipole moment ($\vec{m}$) is always perpendicular to the magnetic field ($\vec{B}$) Hence, $\sin \theta = 1$ always Alternatively: The magnetic field is made radial in a moving coil galvanometer to make the scale linear.	1
It is achieved by using curved magnetic poles. Alternatively:-By using soft iron cylindrical core.	1

10. In a moving coil galvanometer, the deflecting torque τ acting on the coil is related to the current I flowing through it as :

(a) $\tau \propto I^3$

(b) $\tau \propto I^2$

(c) $\tau \propto I$

(d) $\tau \propto \sqrt{I}$

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24. Briefly explain why and how a galvanometer is converted into an ammeter.

2

Explanation of conversion of galvanometer into an ammeter

- Why 1
- How 1

24. Briefly explain why and how a galvanometer is converted into an ammeter.

2

• Due to very high sensitivity Alternatively It has large resistance and hence will change the value of current in circuit. • A galvanometer can be converted into an ammeter of desired range by connecting a shunt of proper value across its coil.	1
	1

Explanation of conversion of galvanometer into an ammeter

- Why 1
- How 1

29. (a) Briefly describe how the current sensitivity of a moving coil galvanometer can be increased.
- (b) A galvanometer shows full scale deflection for current I_g . A resistance R_1 is required to convert it into a voltmeter of range $(0 - V)$ and a resistance R_2 to convert it into a voltmeter of range $(0 - 2V)$. Find the resistance of the galvanometer.

3

- | | |
|---|-----|
| (a) Methods to increase current sensitivity of moving coil galvanometer | (1) |
| (b) Calculation of resistance of galvanometer | (2) |

(a) Current sensitivity of moving coil galvanometer can be increased by (1) increasing number of turns in the coil ($I_s \propto N$) (2) increasing magnetic field strength ($I_s \propto B$) (3) using a material having lesser value of torsional constant $\left(I_s \propto \frac{1}{k} \right)$ (4) increasing area of cross-section of the coil (Note- Any one of the above)	1
(b) $R = \frac{V}{I_g} - G$ For range (0–V) $R_1 = \frac{V}{I_g} - G \text{-----(1)}$ For range (0-2V) $R_2 = \frac{2V}{I_g} - G \text{-----(2)}$ On solving equations (1) and (2) $G = R_2 - 2R_1$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

29. (a) Briefly describe how the current sensitivity of a moving coil galvanometer can be increased.
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(a) Methods to increase current sensitivity of moving coil galvanometer

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(b) Calculation of resistance of galvanometer

(2)

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(2) increasing magnetic field strength ($I_s \propto B$)

(3) using a material having lesser value of torsional constant $\left(I_s \propto \frac{1}{k} \right)$

(4) increasing area of cross-section of the coil

(**Note-** Any one of the above)

1

(b) $R = \frac{V}{I_g} - G$

$\frac{1}{2}$

For range (0–V)

$R_1 = \frac{V}{I_g} - G$ ----- (1)

$\frac{1}{2}$

For range (0-2V)

$R_2 = \frac{2V}{I_g} - G$ ----- (2)

$\frac{1}{2}$

On solving equations (1) and (2)

$G = R_2 - 2R_1$

$\frac{1}{2}$

- 29.** (a) It is not advisable to use a galvanometer as such to measure current directly. Why ?
- (b) Why should the value of resistance connected in parallel to a galvanometer be low ?
- (c) Is the reading shown by an ammeter in a circuit less than or more than the actual value of current flowing in the circuit ? Why ?

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3

(a) Explanation	1
(b) Explanation	1
(c) Answer	$\frac{1}{2}$
Explanation	$\frac{1}{2}$

(a) It will not measure accurate value of current because its high resistance will affect the current in the circuit.	1
(b) To reduce the galvanometer resistance a small resistance is connected in parallel.	1
(c) It is less than the actual value of current because it has some resistance	1

4. A galvanometer of resistance $100\ \Omega$ gives full scale deflection for a current of $1.0\ \text{mA}$. It is converted into an ammeter of range $(0 - 1\text{A})$. The resistance of the ammeter will be close to

(A) $0.1\ \Omega$

(B) $0.8\ \Omega$

(C) $1.0\ \Omega$

(D) $10\ \Omega$

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(A) $0.1\ \Omega$

1. An ammeter and a voltmeter are connected in series to a battery. Their readings are noted as 'A' and 'V' respectively. If a resistor is connected in parallel with the voltmeter, then
- (A) A will increase, V will decrease.
 - (B) A will decrease, V will increase.
 - (C) Both A and V will decrease.
 - (D) Both A and V will increase.

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 - (D) Both A and V will increase.

(A) A will increase, V will decrease

4. A galvanometer shows full scale deflection for a current I_g . If a shunt of resistance S_1 is connected to the galvanometer, it gets converted into an ammeter of range $(0 - I)$. When resistance of the shunt is made S_2 , its range becomes $(0 - 2I)$. Then $\left(\frac{S_1}{S_2}\right)$ is

(A) $\frac{I + I_g}{I - I_g}$

(B) $\frac{I - I_g}{I + I_g}$

(C) $\frac{2I - I_g}{I - I_g}$

(D) $\frac{I - I_g}{2I - I_g}$

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(D) $\frac{I - I_g}{2I - I_g}$

(C) $\frac{2I - I_g}{I - I_g}$

4. A galvanometer of resistance $50\ \Omega$ is converted into a voltmeter of range $(0 - 2\text{V})$ using a resistor of $1.0\ \text{k}\Omega$. If it is to be converted into a voltmeter of range $(0 - 10\ \text{V})$, the resistance required will be
- | | |
|---------------------------|---------------------------|
| (A) $4.8\ \text{k}\Omega$ | (B) $5.0\ \text{k}\Omega$ |
| (C) $5.2\ \text{k}\Omega$ | (D) $5.4\ \text{k}\Omega$ |

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(C) $5.2\ \text{k}\Omega$

8. A galvanometer of resistance $100\ \Omega$ is converted into an ammeter of range $(0 - 1\ \text{A})$ using a resistance of $0.1\ \Omega$. The ammeter will show full scale deflection for a current of about

(A) $0.1\ \text{mA}$

(B) $1\ \text{mA}$

(C) $10\ \text{mA}$

(D) $0.1\ \text{A}$

8. A galvanometer of resistance $100\ \Omega$ is converted into an ammeter of range $(0 - 1\ \text{A})$ using a resistance of $0.1\ \Omega$. The ammeter will show full scale deflection for a current of about

1

(A) $0.1\ \text{mA}$

(B) $1\ \text{mA}$

(C) $10\ \text{mA}$

(D) $0.1\ \text{A}$

(B) 1mA

- (a) (i) (1) What is meant by current sensitivity of a galvanometer ?
Mention the factors on which it depends.
- (2) A galvanometer of resistance G is converted into a voltmeter of range $(0 - V)$ by using a resistance R . Find the resistance, in terms of R and G , required to convert it into a voltmeter of range $\left(0 - \frac{V}{2}\right)$.

(i)(1) Meaning of current sensitivity, mentioning factors	2
(2) Finding the required resistance	$1\frac{1}{2}$
(ii) Finding the induced current	$1\frac{1}{2}$

(i) (1) Current sensitivity of galvanometer is defined as the deflection per unit current.

Alternatively,

$$\frac{\phi}{I} = \frac{NBA}{K}$$

Factors

Number of turns in coil, Magnetic field intensity, Area of coil, Torsional Constant
(Any two)

$$(2) \quad R = \frac{V}{I} - G \quad \text{for (0-V) Range}$$

$$R_1 = \frac{V}{2I} - G \quad \text{for (0-}\frac{V}{2}\text{) Range}$$

$$\frac{V}{I} = R + G$$

$$R_1 = \left(\frac{R+G}{2} \right) - G$$

$$R_1 = \frac{R-G}{2}$$

1

$\frac{1}{2} + \frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (b) (i) What is current sensitivity of a galvanometer ? Show how the current sensitivity of a galvanometer may be increased. “Increasing the current sensitivity of a galvanometer may not necessarily increase its voltage sensitivity.” Explain.
- (ii) A moving coil galvanometer has a resistance $15\ \Omega$ and takes $20\ \text{mA}$ to produce full scale deflection. How can this galvanometer be converted into a voltmeter of range 0 to $100\ \text{V}$?

5

(i) Definition of current sensitivity	1
Showing dependence of current sensitivity & explanation	1+1
(ii) Calculation of resistance	2

(i) Deflection produced per unit current is called its current sensitivity.

$$I_s = \frac{\theta}{I} = \frac{NBA}{K}$$

Current sensitivity can be increased by

- (a) increasing number of turns in coil
 - (b) increasing area of coil in magnetic field
 - (c) decreasing K (Torsional Constant)
- (any one)**

$$V_s = \frac{\theta}{V} = \frac{NBA}{KR}$$

If current sensitivity is increased by increasing number of turns of the coil, the resistance of the galvanometer will also increase. Thus voltage sensitivity may not increase.

(ii) $V = I_G(R + G)$

$$R = \frac{V}{I_G} - G$$

$$= \frac{100}{20 \times 10^{-3}} - 15$$

$$= 5000 - 15$$

$$= 4985 \Omega$$

By connecting 4985Ω in series with galvanometer it is converted to voltmeter of range (0-100V)

1

1

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

5. A galvanometer of resistance $G \Omega$ is converted into an ammeter of range 0 to I A. If the current through the galvanometer is 0.1% of I A, the resistance of the ammeter is :

(A) $\frac{G}{999} \Omega$

(B) $\frac{G}{1000} \Omega$

(C) $\frac{G}{1001} \Omega$

(D) $\frac{G}{100.1} \Omega$

5. A galvanometer of resistance $G \Omega$ is converted into an ammeter of range 0 to $I A$. If the current through the galvanometer is 0.1% of $I A$, the resistance of the ammeter is :

(A) $\frac{G}{999} \Omega$ (B) $\frac{G}{1000} \Omega$ (C) $\frac{G}{1001} \Omega$ (D) $\frac{G}{100.1} \Omega$

(B) $\frac{G}{1000} \Omega$

12. The current sensitivity of a galvanometer does *not* depend on the :

- (A) magnetic field in which the coil is suspended.
- (B) current flowing in the coil.
- (C) torsional constant of the spring.
- (D) area of the coil.

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 - (B) current flowing in the coil.
 - (C) torsional constant of the spring.
 - (D) area of the coil.

(B) Current flowing in the coil.

- 31.** (a) (i) With the help of a labelled diagram, explain the principle of working of a moving coil galvanometer. Write the purpose of using (i) radial magnetic field, and (ii) soft iron core, in it.
- (ii) Define current sensitivity of a galvanometer. “Increasing the current sensitivity may not necessarily increase the voltage sensitivity.” Give reason.

5

(i) Labelled diagram	1
Working principle of moving coil galvanometer	1
Use of (i) Radial magnetic field	$\frac{1}{2}$
(ii) Soft iron core	$\frac{1}{2}$
(ii) Defining current sensitivity	1
Reason	1

Principle:A current carrying coil placed in uniform magnetic field experiences a torque.

(i) Radial magnetic field makes the scale linear
Alternatively: Radial magnetic field provides maximum Torque.

(ii) Use of soft iron core is to increase the strength of magnetic field/
increase sensitivity of the galvanometer.

(ii) **Current sensitivity** is defined as deflection per unit current.
Alternatively:

$$I_s = \frac{\Phi}{I} = \frac{NAB}{k}$$

Voltage sensitivity $V_s = \frac{\Phi}{V} = \left(\frac{NAB}{k}\right) \frac{I}{V} = \left(\frac{NAB}{k}\right) \frac{1}{R}$

Increase in number of turns, increases the current sensitivity and resistance of the galvanometer in the same proportion of current sensitivity therefore Voltage sensitivity remains unchanged.

1

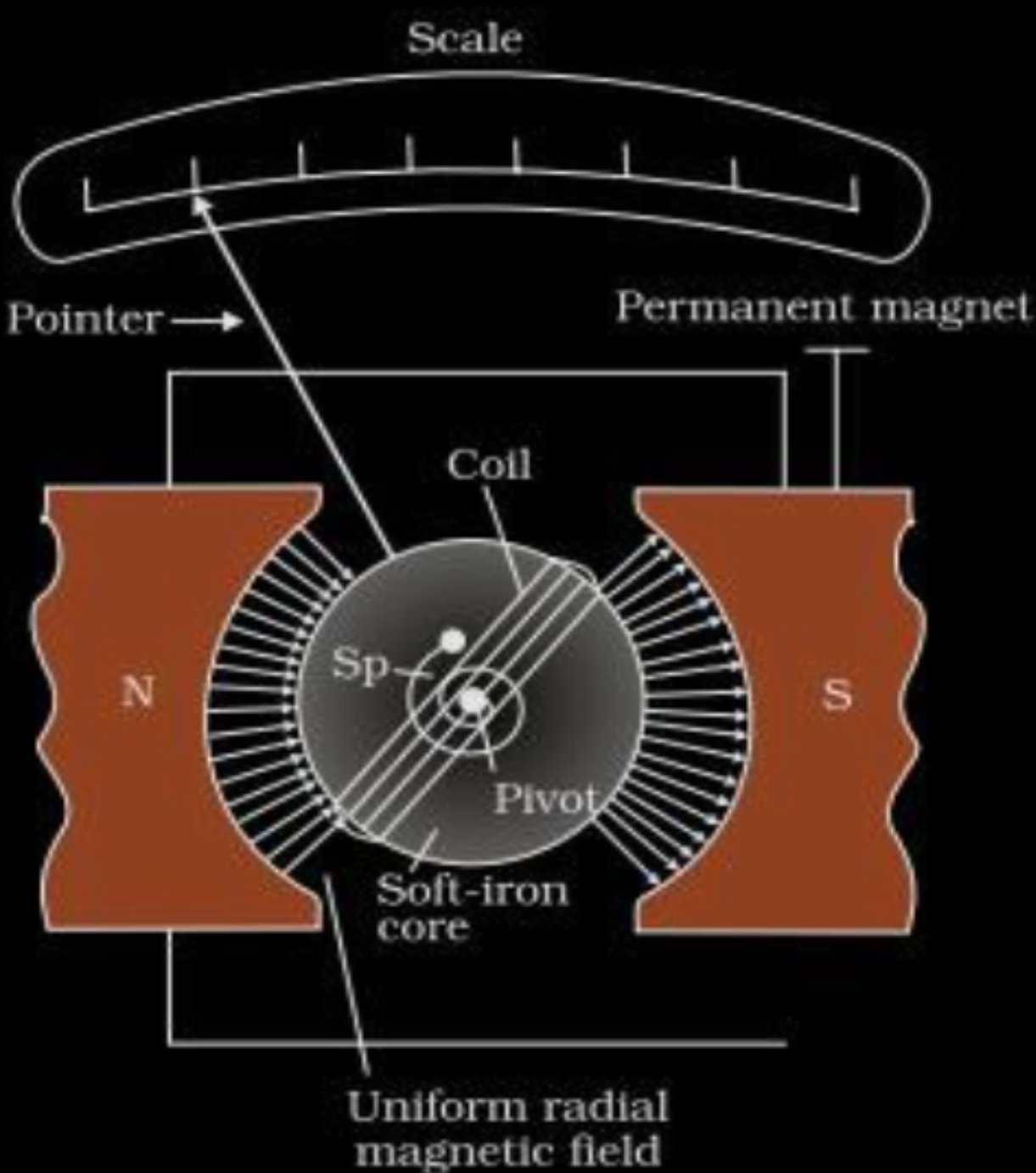
1/2

1/2

1

1/2

1/2



- (b) (i) (I) Write Ampere's circuital law in mathematical form and explain the terms used.
- (II) As the current carrying solenoid is made longer, the magnetic field produced outside it approaches zero. Why ?
- (III) A flexible loop of irregular shape carrying current when located in an external magnetic field, changes to a circular shape. Give reason.
- (ii) A galvanometer of resistance G is converted into a voltmeter to measure up to V volts, by connecting a resistance R_1 in series with the coil. If R_1 is replaced by R_2 , then it can only measure up to $\frac{V}{2}$ volt. Find the value of the resistance R_3 (in terms of R_1 and R_2) needed to convert it into a voltmeter that can read up to $2V$.

(i) (I) Writing Ampere circuital law & explaining the terms.	1
(II) Reason for magnetic field outside long solenoid approaching zero	1
(III) Reason for irregular shaped loop changing to circular loop in uniform magnetic field	1
(ii) Finding the value of Resistance R_3	2

(i) (I) $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_e$ I_e = Total current through the surface B = Magnetic field dl = length of small element	1/2
(II) As length of solenoid increases, it appears like a long cylindrical metal sheet so field outside approaches zero.	1
(III) For a given perimeter, a circle encloses greater area than any other shape, which maximizes the flux.	1
(ii) $R_1 = \frac{V}{I_g} - G \Rightarrow \frac{V}{I_g} = R_1 + G$ -----(1)	1/2
$R_2 = \frac{V}{2I_g} - G \Rightarrow \frac{V}{2I_g} = R_2 + G$ -----(2)	1/2
Solving (1) & (2) $G = R_1 - 2R_2$	1/2
$R_3 = \frac{2V}{I_g} - G$ -----(3)	
Solving using eq (1) & (3) $R_3 = 3R_1 - 2R_2$	1/2

11. A galvanometer can be converted into an ammeter of desired range by connecting a :

- | | |
|----------------------------------|----------------------------------|
| (A) small resistance in series | (B) large resistance in series |
| (C) small resistance in parallel | (D) large resistance in parallel |

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| (A) small resistance in series | (B) large resistance in series |
| (C) small resistance in parallel | (D) large resistance in parallel |

(C) small resistance in parallel

29. A galvanometer is an instrument used to show the direction and strength of the current passing through it. In a galvanometer, a coil placed in a magnetic field experiences a torque and hence gets deflected when a current passes through it. The name is derived from the surname of Italian scientist L. Galvani, who in 1791 discovered that electric current makes a dead frog's leg jerk. A spring attached with the coil provides a counter torque.

In equilibrium, the deflecting torque is balanced by the restoring torque of the spring and we have :

$$NBAI = k\phi$$

where N is the total number of turns in the coil

A is the area of cross-section of each turn

B is the radial magnetic field

k is the torsional constant of the spring

ϕ is the angular deflection of the coil

As the current (I_g) which produces full scale deflection in the galvanometer is very small, the galvanometer cannot as such be used to measure current in electric circuits. A small resistance, called shunt, of a suitable value is connected with the galvanometer to convert it into an ammeter of desired range. By using a higher resistance, a galvanometer can also be converted into a voltmeter.

- (i) (a) A galvanometer is converted into a voltmeter of range $(0 - V)$ by connecting with it, a resistance R_1 . If R_1 is replaced by R_2 , the range becomes $(0 - 2 V)$. The resistance of the galvanometer is :

(A) $(R_2 - 2R_1)$

(B) $(R_2 - R_1)$

(C) $(R_1 + R_2)$

(D) $(R_1 - 2R_2)$

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(A) $(R_2 - 2R_1)$

(B) $(R_2 - R_1)$

(C) $(R_1 + R_2)$

(D) $(R_1 - 2R_2)$

(A) $(R_2 - 2R_1)$

- (b) A current of 5 mA flows through a galvanometer. Its coil has 100 turns, each of area of cross-section 18 cm^2 and is suspended in a magnetic field 0.20 T. The deflecting torque acting on the coil will be :

1

- (A) $3.6 \times 10^{-3} \text{ Nm}$ (B) $1.8 \times 10^{-4} \text{ Nm}$
(C) $2.4 \times 10^{-3} \text{ Nm}$ (D) $1.2 \times 10^{-4} \text{ Nm}$

(B) $1.8 \times 10^{-4} \text{ Nm}$

- (ii) The value of resistance of the ammeter in case (ii) will be : *1*
- (A) $0.20\ \Omega$ (B) $0.24\ \Omega$
- (C) $6.0\ \Omega$ (D) $6.25\ \Omega$

Award 1 mark for this question to all students .

(iii) A galvanometer of resistance $6\ \Omega$ shows full scale deflection for a current of 0.2 A . The value of shunt to be used with this galvanometer to convert it into an ammeter of range $(0 - 5\text{ A})$ is : *1*

(A) $0.25\ \Omega$

(B) $0.30\ \Omega$

(C) $0.50\ \Omega$

(D) $6.0\ \Omega$

(iii) A galvanometer of resistance $6\ \Omega$ shows full scale deflection for a current of 0.2 A . The value of shunt to be used with this galvanometer to convert it into an ammeter of range $(0 - 5\text{ A})$ is : *1*

(A) $0.25\ \Omega$

(B) $0.30\ \Omega$

(C) $0.50\ \Omega$

(D) $6.0\ \Omega$

(A) $0.25\ \Omega$

(iv) The value of the current sensitivity of a galvanometer is given by : *1*

(A) $\frac{k}{NBA}$

(B) $\frac{NBA}{k}$

(C) $\frac{kBA}{N}$

(D) $\frac{kNB}{A}$

(iv) The value of the current sensitivity of a galvanometer is given by : 1

(A) $\frac{k}{NBA}$

(B) $\frac{NBA}{k}$

(C) $\frac{kBA}{N}$

(D) $\frac{kNB}{A}$

(B) $\frac{NBA}{K}$

19. A voltmeter of resistance $1000\ \Omega$ can measure up to 25 V. How will you convert it so that it can read up to 250 V ?

19. A voltmeter of resistance $1000\ \Omega$ can measure up to 25 V . How will you convert it so that it can read up to 250 V ?

2

Conversion of voltmeter to read upto 250V 2	
$I = \frac{V}{R}$ $= \frac{25}{1000}$ $= 25 \times 10^{-3}\text{ A}$	$\frac{1}{2}$
Resistance to be connected to voltmeter $R' = \frac{V'}{I} - R$ $= \frac{250}{25 \times 10^{-3}} - 1000$ $= 9000\ \Omega$	$\frac{1}{2}$ $\frac{1}{2}$
This $9000\ \Omega$ is in series with voltmeter.	$\frac{1}{2}$

16. **Assertion (A)** : The deflection in a galvanometer is directly proportional to the current passing through it. **1**
- Reason (R)** : The coil of a galvanometer is suspended in a uniform radial magnetic field.

16. **Assertion (A)** : The deflection in a galvanometer is directly proportional to the current passing through it. **1**
- Reason (R)** : The coil of a galvanometer is suspended in a uniform radial magnetic field.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion(A).

- (b) To convert a given galvanometer into a voltmeter of ranges 2 V , V and $\frac{\text{V}}{2}$ volt, resistances R_1 , R_2 and R_3 ohm respectively, are required to be connected in series with the galvanometer. Obtain the relationship between R_1 , R_2 and R_3 .

b)

$$R = \frac{V}{i_g} - G$$

1/2

$$R_1 = \frac{2V}{i_g} - G = R_o - G$$

$$R_1 + G = 2R_o$$

1/2

$$\left[\text{Where } R_o = \frac{V}{i_g} \right]$$

Similarly

$$R_2 + G = R_o$$

$$R_3 + G = R_o/2$$

1/2

From the above equations,

$$R_1 - R_2 = 2(R_2 - R_3)$$

$$R_1 - 3R_2 + 2R_3 = 0$$

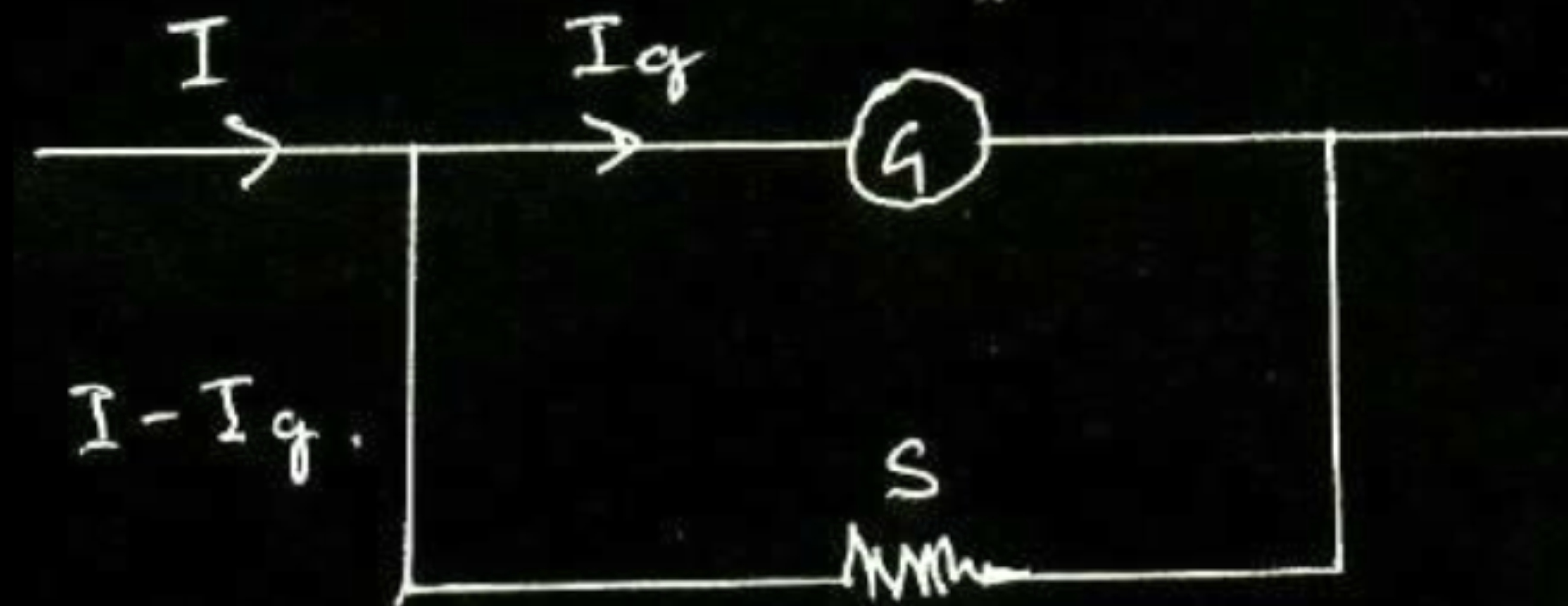
1/2

- 22.** An ammeter of resistance $0.8\ \Omega$ can measure a current up to 1.0 A . Find the value of shunt resistance required to convert this ammeter to measure a current up to 5.0 A .

22. An ammeter of resistance 0.8Ω can measure a current up to 1.0 A . Find the value of shunt resistance required to convert this ammeter to measure a current up to 5.0 A .

2

Diagram	$\frac{1}{2}$
Formula	$\frac{1}{2}$
Calculation of value of shunt	1



$$\begin{aligned}
 I_g &= 1 \text{ A} \\
 \text{Resistance of ammeter, } R_A &= 0.8 \Omega \\
 I_g R_A &= (I - I_g) S \\
 \Rightarrow 1 \times 0.8 &= (5 - 1) S \\
 \Rightarrow S &= 0.2 \Omega
 \end{aligned}$$

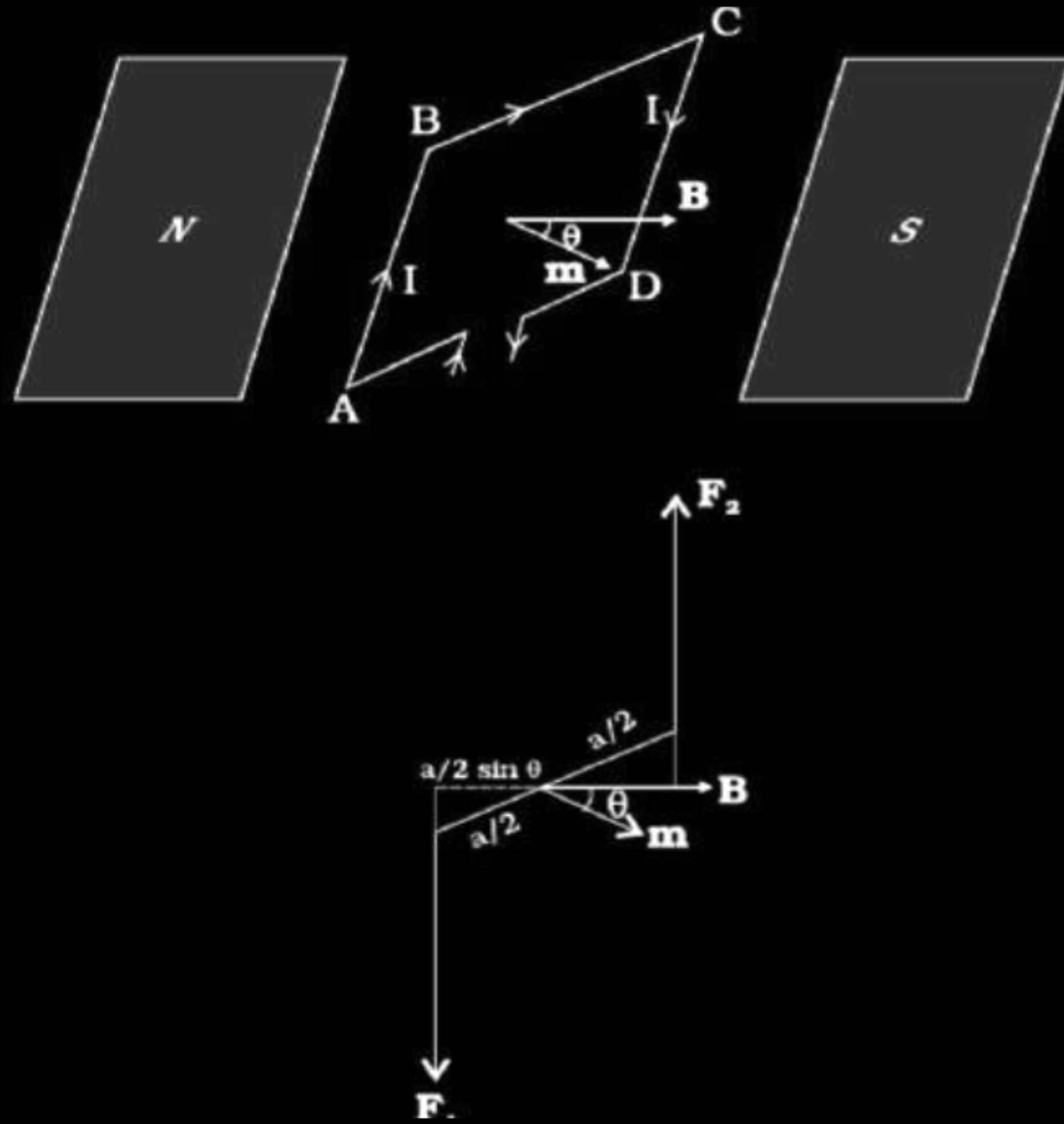
$\frac{1}{2}$

1

$\frac{1}{2}$

- (a) Obtain the expression for the deflecting torque acting on the current carrying rectangular coil of a galvanometer in a uniform magnetic field. Why is a radial magnetic field employed in the moving coil galvanometer ?

(a)

 $\frac{1}{2}$

Arms AD and BC experience no net force whereas arm AB and CD experience forces which constitute torque
 $F_1 = F_2 = I b B$

Therefore magnitude of torque

$$\tau = \left(F_1 \frac{a}{2} + F_2 \frac{a}{2} \right) \sin \theta$$

$$\tau = I(ab) \sin \theta$$

where $ab = A$ (area of the loop)

$$\Rightarrow \tau = IAB \sin \theta$$

for N number of turns

$$\tau = NIAB \sin \theta$$

$$\tau = \vec{M} \times \vec{B}$$

Where magnetic moment $M = NIA$

Galvanometer has Radial magnetic field to increase the field strength and to make torque independent of orientation θ / it maximise the torque

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

1

- (a) Define current sensitivity of a galvanometer. Write its expression.
- (b) A galvanometer has resistance G and shows full scale deflection for current I_g .
- (i) How can it be converted into an ammeter to measure current up to I_0 ($I_0 > I_g$) ?
- (ii) What is the effective resistance of this ammeter ?

3

a) Definition and expression	1 mark
b) Conversion of Galvanometer	
(i) into ammeter	1 mark
(ii) Effective resistance	1 mark

a) Deflection per unit current

$$I_s = \frac{\theta}{I} = \frac{BNA}{K}$$

b) (i) By connecting a low resistance (R_s) in parallel to galvanometer such that

$$(I_0 - I_g)R_s = I_g G$$

(ii) effective resistance

$$\frac{1}{R_A} = \frac{1}{R_s} + \frac{1}{G} = \frac{G + R_s}{R_s G}$$

$$\therefore R_A = \frac{R_s G}{G + R_s}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

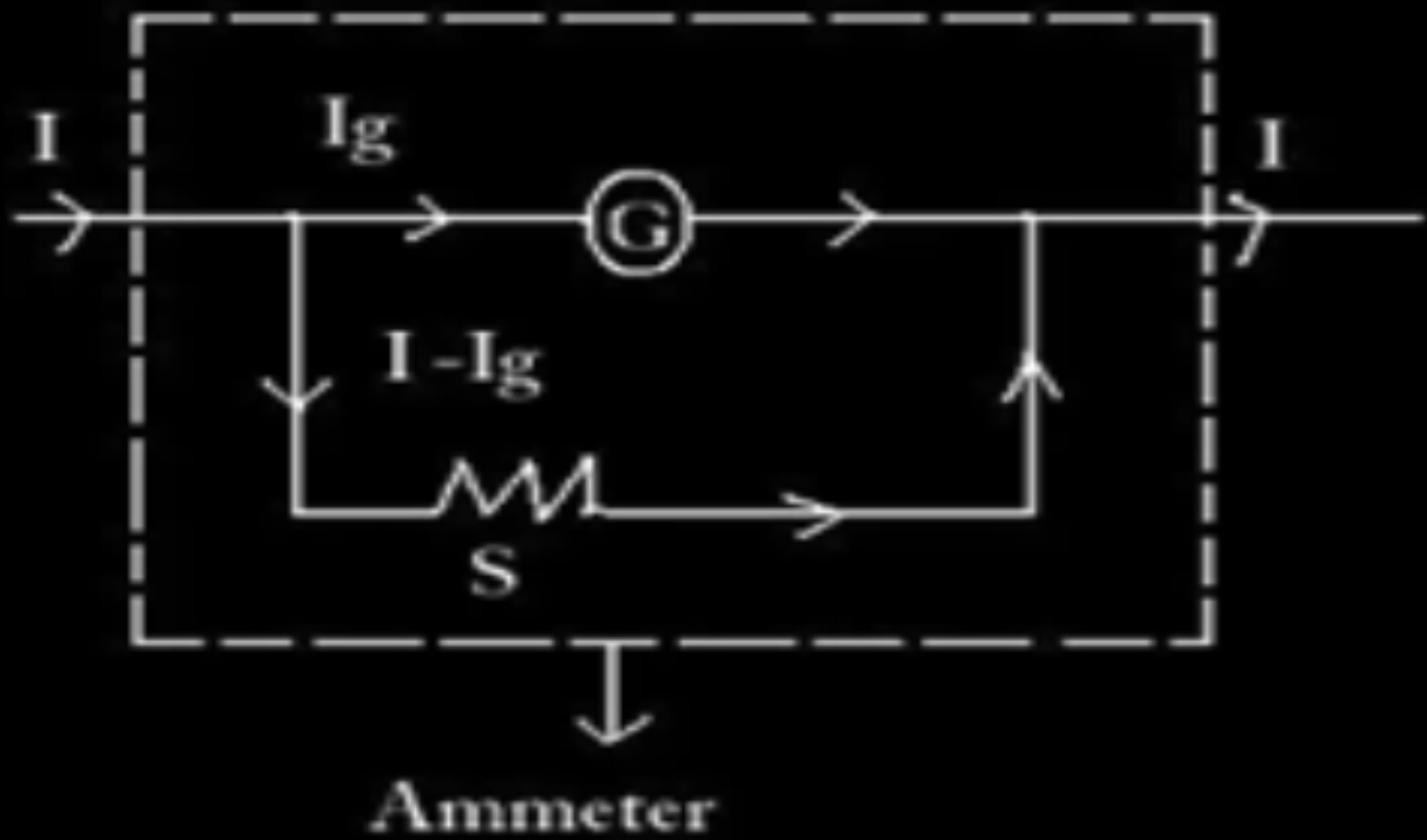
State the principle of a moving coil galvanometer. Explain its working and obtain the expression for the deflection produced due to the current passed through the coil. Define current sensitivity.

OR

Explain how a galvanometer can be converted into an ammeter of a given range. Derive an expression for shunt resistance and current for full scale deflection. Find the effective resistance of the ammeter.

Principle -	1
Working & expression for deflection -	1
Current Sensitivity -	1

Conversion of galvanometer into ammeter -	1
Expression for shunt resistance -	1
Effective resistance of ammeter -	1

Principle - A current carrying coil experiences a torque in a magnetic field.	1
Working - When current is passed through the coil torque produced is $\tau = NIAB \sin \theta = NIAB \quad (\theta=90^\circ)$	1/2
Restoring torque $\tau' = k\phi$ At equilibrium $\tau = \tau'$ $NIAB = k\phi$ $\phi = \left(\frac{NAB}{k}\right)I$	1/2
Current sensitivity: $\frac{\phi}{I} = \frac{NAB}{k}$	1
[Alternatively: It is deflection per unit current]	
A galvanometer is converted into an ammeter when a suitable shunt resistance is connected in parallel with the galvanometer.	1/2
	1/2
[Award full one mark even if no figure drawn or only the figure is drawn]	

$G I_g = (I - I_g) S$	1/2
$S = \left(\frac{I_g}{I - I_g}\right) G$	1/2
$R_{eq} = \frac{GS}{G + S}$	1

- (a) Briefly explain how a galvanometer is converted into an ammeter.
- (b) A galvanometer coil has a resistance of $15\ \Omega$ and it shows full scale deflection for a current of $4\ \text{mA}$. Convert it into an ammeter of range 0 to $6\ \text{A}$.

i) Conversion of a galvanometer into an ammeter	$1\frac{1}{2}$
ii) Formula for shunt	$\frac{1}{2}$
iii) Calculation and Result	1

a)	By connecting a small resistance called shunt (S) in parallel to coil of the galvanometer The value of S is related to the maximum current (I) to be measured as $S = I_g G / (I - I_g)$. (Note : If the student just draws the diagram, full marks may be awarded).	1½
b)	$G = 15 \Omega$ $I_g = 4 \times 10^{-3} \text{ A}$ $I = 6 \text{ A}$ $I_g G = (I - I_g) S$ $S = \frac{I_g G}{I - I_g} = \frac{4 \times 10^{-3} \times 15}{6 - 4 \times 10^{-3}}$ $= 0.01 \Omega$	 ½ ½ ½

- (a) Briefly explain how a galvanometer is converted into a voltmeter.
- (b) A voltmeter of a certain range is constructed by connecting a resistance of $980\ \Omega$ in series with a galvanometer. When the resistance of $470\ \Omega$ is connected in series, the range gets halved. Find the resistance of the galvanometer.

i) Conversion of a galvanometer into a voltmeter	$1\frac{1}{2}$
ii) Formula	$\frac{1}{2}$
iii) Calculation and results	1

a) A galvanometer may be converted into voltmeter by connecting a high value resistance R in series with coil of the galvanometer. The value of (R) is related to the maximum voltage (V) to be measured as $R = \frac{V}{I_g} - G$ Note - Award full marks if a student just draws the labelled diagram.	1½
b) $I_g = \frac{V}{R_g + R}$ $\frac{V}{R_g + 980} = \frac{V}{2(R_g + 470)}$ $2R_g + 940 = R_g + 980$ $R_g = 40\Omega$	½ ½ ½

How is a galvanometer converted into a voltmeter and an ammeter ? Draw the relevant diagrams and find the resistance of the arrangement in each case. Take resistance of galvanometer as G .

How galvanometer is converted in to a voltmeter and an Ammeter	$\frac{1}{2} + \frac{1}{2}$
Diagram for conversion of galvanometer into a voltmeter and an Ammeter.	$\frac{1}{2} + \frac{1}{2}$
Resistance of each arrangement	$\frac{1}{2} + \frac{1}{2}$

A galvanometer is converted into a voltmeter by connecting a high resistance 'R' in series with it.

1/2

A galvanometer is converted into an ammeter by connecting a small resistance (called shunt) in parallel with it.

1/2

Resistance of voltmeter, $R_V = G + R$

1/2

Resistance for Ammeter, $R_A = \frac{G r_s}{G + r_s}$

1/2

